The drops that did not fall.

01 Can a leaf get wet without rain?



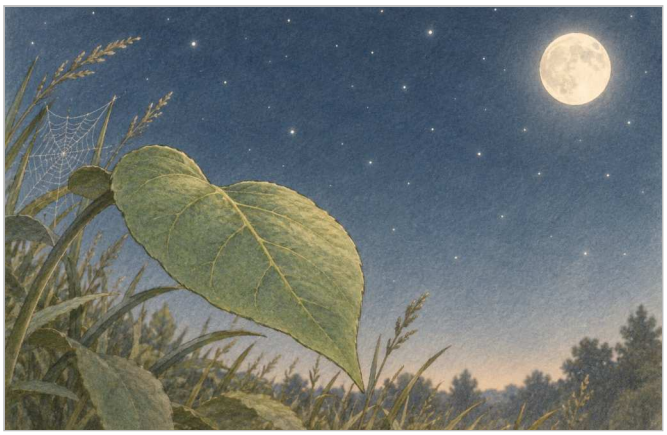
An imagined close look. Drops are enlarged; the picture is not counting data.

Look closer. Look at the drops on the leaf and the nearby spider web.

A little explanation. Yes. Air holds invisible water vapor. Dew forms right on a cool surface when that vapor condenses into liquid drops. It does not fall from the sky like rain.

Try this. Point to where the drops sit. Imagine a drop growing there, instead of falling down.

02 How does a leaf cool during the night?



An imagined close look. Drops are enlarged; the picture is not counting data.

Look closer. The leaf faces a wide, clear sky. The grass is still.

A little explanation. Surfaces give off heat to the sky. On a clear, calm night, a leaf can lose heat and get colder than the air around it. Clouds and wind can slow that cooling.

Try this. Trace the leaf's edge. Say the first command: "Cool the leaf."

03 Where was the water before we saw drops?



An imagined close look. Drops are enlarged; the picture is not counting data.

Look closer. The air above the leaf looks clear. New beads cling to the leaf.

A little explanation. The water was invisible vapor in the air. Air touching the cool leaf cools too. At its dew point, it is cool enough that it cannot keep all its water invisible. Some vapor condenses into liquid dew.

Try this. Imagine clear air touching this leaf. Use "condense" to tell what changed.

04 Where do the drops go after sunrise?



An imagined close look. Drops are enlarged; the picture is not counting data.

Look closer. Sunlight reaches the leaf. Fewer drops remain.

A little explanation. As the sun warms the leaf, liquid water evaporates back into invisible vapor. The water changes form; it does not vanish forever. A cold glass can collect drops on its outside for the same condensation reason: that water comes from the surrounding air.

Try this. Say two commands in order: "Warm the leaf. Watch the drops shrink."